IOT- BASED WATER SUPPLY TIME SCHEDULER FOR VILLAGES

A project submitted to the Bharathidasan University
in partial fulfillment of the requirements
for the award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted by

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DEPARTMENT OF COMPUTER APPLICATIONS

BISHOP HEBER COLLEGE (AUTONOMOUS)

"Nationally Reaccredited at A++ Grade with a CGPA of 3.69 out of 4 in NAAC IV Cycle"

Recognized by UGC as "College of Excellence"
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TIRUCHIRAPPALLI-620 017

OCTOBER – 2025

DECLARATION

I hereby declare that the project work presented is originally done by me under the guidance of **Dr. K. MAKESH BABU , MCA., M.S(SS)., M.Phil., PGDBA., Ph.D., Assistant Professor, UG Department of Computer Applications, Bishop Heber College (Autonomous), Tiruchirappalli-620 017** and has not been included in any other thesis/project submitted for any other degree.

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This is to certify that the project work entitled **"IOT-BASED WATER SUPPLY TIME SCHEDULER FOR VILLAGES "** is a bonafide record work done by MILTON S(235113558), MOHAMED ASIC N (235113561), PRASANNAVASAN R (235113564), KUMARAN N M (235113565), in partial fulfillment of the requirements for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS during the period 2023 - 2026.

Place: Trichy **Signature of the Guide**

UG DEPARTMENT OF COMPUTER APPLICATIONS

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TIRUCHIRAPPALLI – 620 017

Date:

Course Title: Project **Course Code: U23CA5PJ**

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1. ABSTRACT

Water scarcity and improper distribution remain pressing issues in rural villages, where limited infrastructure often leads to wastage, unequal access, and lack of timely information about water availability. Conventional water supply systems in these areas operate without efficient monitoring or scheduling, causing tanks to overflow or remain underutilized, which further intensifies the problem. To address these challenges, this project proposes a Smart Water Supply Time Scheduler using Internet of Things (IoT) technology. The system is designed to continuously monitor the water flow and tank level using sensors, ensuring real-time tracking of the supply process. Whenever water enters the tank, the system automatically sends notification messages to the villagers, informing them about availability, thereby allowing households to plan their usage more efficiently. Additionally, the system detects overflow conditions and generates alerts to prevent wastage, ensuring sustainable usage of resources. The time scheduling feature enables regulated water distribution across households, improving fairness and reliability. By integrating IoT-based communication and sensor technologies, the proposed system enhances transparency, reduces human intervention, and promotes efficient management of rural water supply. Ultimately, this approach addresses the current problems of unregulated water usage and supply mismanagement, contributing to sustainable rural development.

INTRODUCTION

2. INTRODUCTION

Water is one of the most essential resources for sustaining life, economic development, and environmental balance. However, the rapid growth of population, urbanization, and industrialization has placed unprecedented pressure on the global water supply. In many rural and urban areas, people still face difficulties in accessing sufficient and clean water for their daily needs. Traditional methods of water distribution and management are often inefficient, leading to challenges such as water scarcity, leakage, wastage, and unequal distribution. These issues are more critical in villages and semi-urban areas, where the infrastructure is underdeveloped and manual supervision of water resources becomes highly challenging. In this context, the need for a modern, intelligent, and automated approach to water management has become a pressing necessity. The concept of Smart Water Supply Management has emerged as a solution that integrates advanced technologies like the Internet of Things (IoT), sensors, and communication networks to provide a reliable, transparent, and efficient water distribution system. By leveraging real-time monitoring, data collection, and automation, smart systems aim to optimize water supply, reduce wastage, and ensure sustainable use of this vital resource.

Fig1.1 Water Management

The conventional water management systems in many regions often rely on manual control and observation. For instance, village water tanks are filled at fixed intervals without consideration of real-time demand or current tank levels, which can cause either water shortages or unnecessary overflow and wastage. In urban contexts, water leakage from pipelines goes unnoticed for extended periods, resulting in huge losses of treated water. Moreover, unequal supply between regions creates disputes and dissatisfaction among residents. These limitations reveal the inefficiency of manual systems in addressing the growing demand for water management. The advent of IoT-based smart water supply management introduces a paradigm shift, enabling continuous monitoring of water levels, flow rates, and distribution patterns. With sensor-based monitoring and automated control systems, it becomes possible to regulate the water supply based on actual demand and availability. For example, sensors installed in overhead tanks can measure water levels and send notifications to local authorities or end-users when the tank is full or nearing depletion. This prevents both overflow and unexpected shortages, ensuring that water is used judiciously and distributed fairly.

Another major aspect of smart water supply management is its ability to provide transparency and accountability. Traditional systems often suffer from a lack of data, which makes it difficult for authorities to identify where and how water is being consumed or wasted. With smart solutions, data regarding water consumption, leakage detection, and flow rates can be collected and analyzed in real time. This empowers decision-makers to take corrective actions promptly, such as repairing leaks, reallocating water to underserved areas, or scheduling supply based on actual demand. Moreover, by integrating mobile applications and SMS alerts, end-users can also be informed about water availability, tank status, or supply schedules. Such real-time communication bridges the gap between water authorities and citizens, fostering trust and participation in water

conservation. Additionally, data analytics can help in predicting future consumption trends, thereby assisting policymakers in planning for sustainable water resource allocation.

Sustainability and conservation are at the heart of smart water supply management. Water scarcity is becoming a global challenge due to climate change, overuse of freshwater resources, and uneven rainfall distribution. Countries worldwide are focusing on solutions that reduce wastage and promote efficient utilization. Smart systems not only help prevent losses due to leakage and overflow but also encourage consumers to use water responsibly. For instance, smart meters can be installed in households or communities to record usage patterns, providing insights into consumption behavior. This information can be used to create awareness campaigns or even implement incentive-based conservation programs. By promoting responsible water usage, smart management systems contribute to long-term sustainability while also ensuring that sufficient resources remain available for future generations. Furthermore, these systems are scalable and adaptable, meaning they can be implemented in both small rural villages and large metropolitan cities, thus making them universally relevant.

Technological advancements in IoT, cloud computing, and wireless communication have made the deployment of smart water systems both feasible and cost-effective. IoT-enabled sensors are capable of transmitting real-time data over long distances with minimal power consumption, which is particularly beneficial for rural areas with limited infrastructure. Cloud-based platforms allow large volumes of data to be stored, processed, and analyzed, providing actionable insights to authorities and stakeholders. Additionally, integration with mobile networks ensures that even end-users in remote areas receive timely notifications regarding their water supply. The automation of pumps and valves based on real-time tank levels or pre-defined thresholds eliminates the

need for constant human intervention, thereby reducing labor costs and operational inefficiencies. These technological features make smart water supply management not only efficient but also economically viable in the long run.

In the importance of smart water supply management lies in its potential to address some of the most pressing challenges related to water distribution and utilization. By combining IoT-based monitoring, automation, and real-time communication, such systems can minimize wastage, prevent shortages, and ensure equitable distribution of water resources. They also promote sustainability through conservation practices and provide transparency by making data accessible to both authorities and end-users. As the demand for water continues to rise globally, the adoption of smart water management systems is no longer an option but a necessity. Whether in rural villages facing unreliable water supply or urban cities struggling with leakage and inefficiencies, the application of smart technologies offers a path toward a sustainable and efficient water future. Ultimately, Smart Water Supply Management represents a transformative step in how societies interact with and preserve one of their most critical resources.

EXISTING SYSTEM AND PROPOSED SYSTEM

3. EXISTING SYSTEM AND PROPOSED SYSTEM

3.1 EXISTING SYSTEM

The water management system in most areas is still dependent on traditional and manual practices, which are inefficient in addressing the growing demand for water. In rural regions, water tanks are usually filled at fixed times without considering real-time demand, leading to frequent overflows or shortages. Since there is no automated mechanism, human supervision is required, which is often irregular and prone to error. In urban areas, water is supplied on pre-decided schedules, but leakages in pipelines go unnoticed for long periods, resulting in significant water wastage. Consumers usually do not receive any alerts or prior information about the availability of water, which causes inconvenience in daily life. Mechanical water meters, where available, require manual readings and cannot provide accurate real-time data. Authorities also lack access to reliable information about consumption patterns, making it difficult to balance demand and supply effectively. Unequal water distribution between different localities is another common issue, with some regions receiving more water while others face scarcity. Transparency and communication between supply authorities and end-users are very limited in the existing system. Problems like leakage, wastage, and overflow are identified only after serious damage has occurred. Moreover, there is no integration of sustainability practices, and water conservation is rarely encouraged through existing methods. Overall, the current system provides basic water supply but fails to ensure efficiency, accountability, and sustainability.

3.1.1 DISADVANTAGES

- Manual dependency
- High wastage
- Unequal distribution

- No transparency
- Delayed detection

3.2 PROPOSED SYSTEM

The proposed system introduces a Smart Water Supply Time Scheduler using IoT to overcome the challenges of unregulated water distribution and wastage in villages. The system consists of water flow sensors and level sensors installed in overhead or underground storage tanks to monitor real-time water availability. When water is supplied into the tank, the system detects the inflow and automatically sends notification messages to villagers through mobile communication modules, ensuring that people are informed about water availability without delay. Similarly, when the tank reaches its maximum capacity, the system generates an overflow alert to prevent water loss. A microcontroller acts as the central unit, processing the data from sensors and coordinating communication with the users. Additionally, a scheduling module is integrated to manage the timing of water supply across different households, promoting fair and systematic distribution. The IoT platform ensures that the data is logged, analyzed, and accessible for monitoring by local authorities to maintain transparency. This system reduces manual intervention, minimizes wastage, and ensures that villagers receive timely updates, thereby improving the overall efficiency and sustainability of the rural water supply network.

3.2.1 ADVANTAGES

- Prevents water wastage through overflow alerts.
- Ensures fair distribution with scheduled supply.
- Provides real-time updates to villagers.
- Reduces manual monitoring efforts.
- Improves sustainability of rural water management.

HARDWARE REQUIREMENT AND SOFTWARE REQUIREMENT

4. HARDWARE REQUIREMENT AND SOFTWARE REQUIREMENT

4.1 Hardware Requirements

- Flow sensor
- Arduino Nano
- Buzzer
- Relay
- GSM
- Power
- LCD display

4.2Software Requirement

- Arduino IDE

4.3Hardware Description

Flow Sensor

A flow sensor is a measuring device that detects and quantifies the movement of a liquid or gas through a pipe or channel. It is one of the most important components in process industries, water distribution systems, and smart monitoring applications because it ensures accurate measurement of flow rate and overall system efficiency. Flow sensors can measure parameters such as volume, velocity, or mass of the fluid depending on the type of sensor and its design. These sensors generally consist of a sensing element that interacts with the moving fluid, and an electronic circuit that converts the detected signals into readable outputs, either analog or digital. The most commonly used types of flow sensors include turbine flow sensors, ultrasonic sensors, electromagnetic

flow meters, differential pressure-based sensors, and thermal mass flow sensors. Each type is suitable for specific conditions; for instance, electromagnetic sensors are ideal for conductive liquids like water, while ultrasonic sensors are effective in applications where non-intrusive, clamp-on measurement is required.

Flow sensors are widely used in irrigation systems, industrial fluid management, automotive fuel monitoring, HVAC systems, and medical devices such as ventilators and infusion pumps. In water management, they are applied to detect consumption patterns, prevent wastage, and give real-time updates when leakage occurs. When integrated with IoT modules, flow sensors can send data wirelessly to cloud platforms, allowing remote monitoring and predictive analytics for maintenance. This has become especially useful in smart city applications and rural water distribution systems. The accuracy and reliability of flow sensors help industries maintain safety standards and reduce operational costs.

Another important aspect of flow sensors is their role in environmental monitoring. By keeping track of wastewater discharge or chemical effluents, industries can ensure compliance with pollution control regulations. In medical applications, precise flow measurement is critical for patient safety, as even small deviations can cause health risks. Flow sensors also find applications in power plants for monitoring cooling water and in fuel supply chains for measuring petroleum flow. Modern advancements have made sensors more compact, energy-efficient, and compatible with microcontrollers such as Arduino, Raspberry Pi, and industrial PLCs. In conclusion, flow sensors serve as the backbone of fluid monitoring systems, enabling efficient management, cost savings, and sustainable resource utilization.

RELAY

A relay is an electrically operated switch that allows a low-power circuit to control a higher-power circuit safely and effectively. It plays a key role in electrical and electronic systems where isolation, protection, and control are required. Relays are mainly of two types: electromechanical relays (EMR) and solid-state relays (SSR). In electromechanical relays, a coil energizes and creates a magnetic field, which pulls a set of contacts to open or close the circuit. Solid-state relays, on the other hand, use semiconductor devices for switching and have no moving parts, offering faster response and longer life. Relays are further classified into types such as thermal relays, reed relays, latching relays, and time-delay relays depending on their design and function.

The major application of relays is in automation and protection systems. In industrial automation, relays are used for motor starters, machine interlocking, and process sequencing. In power systems, protective relays detect faults like overcurrent, under-voltage, or overload, and disconnect faulty sections to prevent damage to equipment. In the automotive sector, relays are used to control headlights, ignition, air conditioning, and other electronic subsystems. In household appliances, they are employed in washing machines, microwave ovens, refrigerators, and air conditioners for safe switching. Solid-state relays are widely used in applications that demand noise-free, high-speed operation, such as telecommunications and digital control systems.

One of the most important functions of a relay is isolation, where the control circuit is electrically separated from the load circuit. This prevents hazards and protects sensitive electronic components. Relays can be designed for single-pole or multiple-pole switching, allowing them to control multiple circuits simultaneously. Their contact ratings vary based on application, from small signal switching in electronic devices to heavy-duty industrial loads. With

technological advancements, modern relays are more compact, energy-efficient, and capable of handling high switching frequencies without wear and tear.

Relays are also central to safety and emergency systems. For instance, they are used in elevators, fire alarm systems, and railway signaling systems where reliability and quick response are critical. In renewable energy applications such as solar inverters and wind turbines, relays ensure safe disconnection during faults and contribute to efficient power management. In automation and IoT-based smart control systems, relays are often interfaced with microcontrollers and wireless modules to enable remote operation of devices. Thus, relays are not just switches but intelligent control elements that combine safety, flexibility, and efficiency in modern electrical systems.

ARDUINO NANO

One kind of microcontroller board is the Arduino Nano, which is created by Arduino.cc. An Atmega328 microcontroller can be used to construct it. The Arduino UNO also makes use of this microcontroller. This versatile board comes in a compact size and offers an extensive range of uses. The Arduino Mega, Arduino Pro Mini, Arduino UNO, Arduino YUN, Arduino Lilypad, Arduino Leonardo, and Arduino Due are the principal other Arduino boards. AVR Development Board, PIC Development Board, Raspberry Pi, Intel Edison, MSP430 Launchpad, and ESP32 board are some additional development boards. This board is similar to an Arduino Duemilanove board in that it offers a lot of features. The packaging for this Nano board is distinct, though. It lacks a DC jack, which allows the power supply to be supplied via a tiny USB port instead of being directly attached to pins like VCC and GND. A small USB connector on the board allows you to supply 6 to 20 volts to this board

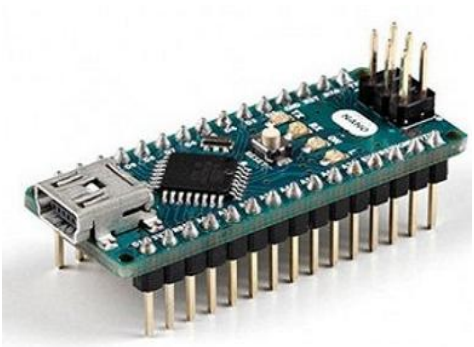

Fig Arduino Nano

- **Started With The Arduino Nano**

The Arduino Nano is a small, complete, and breadboard-friendly board based on the ATmega328P. It offers the same connectivity and specs of the UNO board in a smaller form factor.

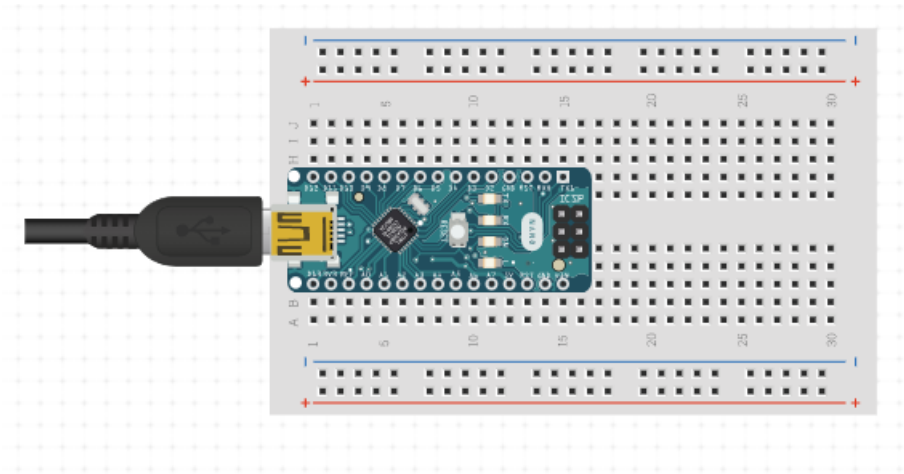

Fig Arduino NANO Interface

The Arduino Nano is programmed using the Arduino Software (IDE), our Integrated Development Environment common to all our boards and running both online and offline. For more information on how to get started with the Arduino Software visit the Getting Started page.

Arduino Nano on the Arduino Desktop IDE

If you want to program your Arduino Nano while offline you need to install the Arduino Desktop IDE To connect the Arduino Nano to your computer, you'll need a Mini-B USB cable. This also provides power to the board, as indicated by the blue LED (which is on the bottom of the Arduino Nano 2.x and the top of the Arduino Nano 3.0).

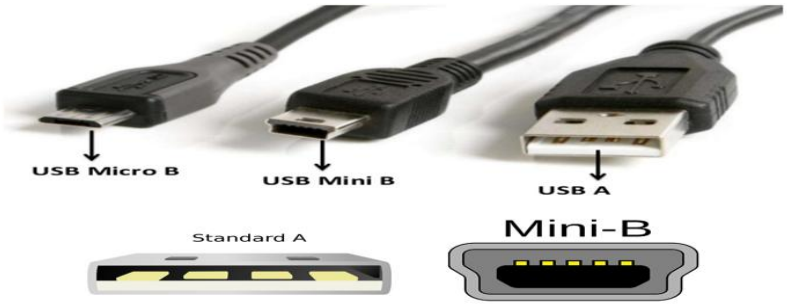

Fig NANO Interfacing USB Types of Port

Open your first sketch

- Open the LED blink example sketch: File > Examples > 01.Basics > Blink.

Select your board type and port

- Select Tools > Board > Arduino AVR Boards > Arduino Nano.

NOTE: We have updated the Nano board with a fresh bootloader. Boards sold by us from January 2018 have this new bootloader, while boards manufactured before that date have the old bootloader. First, check that **Tools > Board > Boards Manager shows you have the Arduino AVR Boards** 1.16.21 or later installed. Then, to program the NEW Arduino NANO boards you need to choose **Tools > Processor > ATmega328P**. To program old boards you need to choose **Tools > Processor > ATmega328P (Old Bootloader)**. If you get an error while uploading or you are not sure which bootloader you have, try each Tools > Processor menu option until your board gets properly programmed.

GSM (Global System for Mobile Communications)

GSM (Global System for Mobile Communications) is a widely adopted standard for mobile communications, enabling cellular networks to provide services such as voice calls, text messages, and data transmission. Developed in the 1980s and initially deployed in Europe, GSM has become the global standard for mobile phone systems, offering a reliable, scalable, and secure communication platform. The GSM network operates on two main components: the mobile station (MS), which includes the user's mobile device, and the base station subsystem (BSS), which connects the mobile device to the network.

Fig GSM

The GSM system relies on a digital transmission method, using Time Division Multiple Access (TDMA) to allocate time slots for multiple users within the same frequency channel. This method allows for efficient use of available spectrum and supports high capacity in mobile communication systems. GSM also incorporates encryption protocols to ensure the confidentiality and integrity of communication, making it a secure platform for voice and data exchanges.

A key feature of GSM is the SIM (Subscriber Identity Module) card, which holds user-specific data such as phone numbers, contacts, and security information. This card makes it easy for users to switch between devices without losing personal information. Additionally, GSM supports services such as SMS (Short Message Service), enabling text communication between mobile devices, and GPRS (General Packet Radio Service) for basic data services, providing internet access and multimedia messaging capabilities.

Despite the rise of newer communication technologies such as 4G and 5G, GSM remains widely used in many parts of the world, especially in rural or developing areas where the infrastructure for more advanced systems may not be readily available. It continues to serve as a reliable communication system for voice, text, and basic data services.

LCD (LIQUID CRYSTAL DISPLAY)

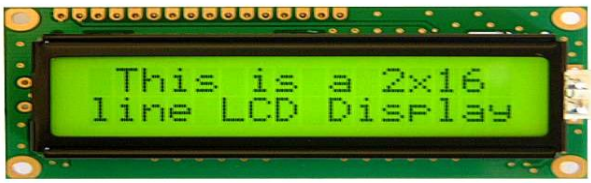

Fig LCD Display

LCD Display

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits.

A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD, The data register stores the data to be displayed on the LCD

The data is the ASCII value of the character to be displayed on the LCD.Liquid crystal displays are used for display of numeric and alphanumeric character in dot matrix and segmental displays. The two liquid crystal materials which are commonly used in display technology are nematic and cholesteric whose schematic arrangement of molecules is shown in fig. The most popular liquid crystal structure is the Nematic Liquid Crystal (NLC). In this all the molecules align themselves approximately parallel to a unique axis (director), while retaining the complete translational freedom. The liquid is normally transparent, but if subjected to a strong electric field, disruption of the well ordered crystal structure takes place causing the liquid to polarize and turn opaque. The removal of the applied electric field allows the crystal structure to regain its original form and the materials become transparent.

Based on the construction, LCD's are classified into two types. They are,

(i) Dynamic scattering type

(ii) Field effect type.

Dynamic scattering type

The construction of the dynamic scattering liquid crystal cell is shown in the fig. The display consists of two glass plates, each coated with tin oxide (SnO_2) on the inside with transparent electrodes separated by a liquid crystal layer,5μA to 50 μA thick. The oxide coating on the front sheet is etched to produce a single or multi- segment pattern of characters, with each segment

properly insulated from each other. A weak electric field applied to liquid crystal tends to align molecule in the direction of the field. As soon as the voltage exceeds certain threshold value, the domain structure collapses and the appearance is changed. As the voltage grows further, the flow becomes turbulent and the substance turns optically homogenous. In this disordered state, the liquid crystal scatters light.

Thus, when the liquid is not activated, it is transparent. When the liquid is activated, the molecular turbulence causes light to be scattered in all directions and the cell appears bright. This phenomenon is called dynamic scattering

Field effect type

The construction of the field effect LCD display is similar to that of the dynamic scattering type, with the expectation that two thin polarizing optical filters are placed at the inside of each glass sheet.

The LCD material is of twisted nematic type which twists the light (change in direction of polarization) passing through the cell when the latter is not energized. This allows light to pass through the optical filters and the cell appears bright. When the cell is energized, no twisting of light takes place and the cell appears dull.

Liquid crystal cells are of two types (i) transitive type (ii) reflective type. In the transitive type cell both glass sheets are transparent so that the light from the rear source is scattered in the forward direction when the cell is activated. The reflecting type cell has a reflecting surface on one side of the glass sheet. The incident light on the front surface of the cell is dynamically scattered by an activated cell. Both types of cells appear quite bright when activated even under ambient light conditions.

Liquid crystals consume small amount of energy, in a seven segment display the current drawn is about 25 µA for dynamic scattering cells and 300 µA .

for field effect cells LCD's require ac voltage supply. A typical voltage supply to dynamic scattering LCD's are normally used for seven-segmental displays.

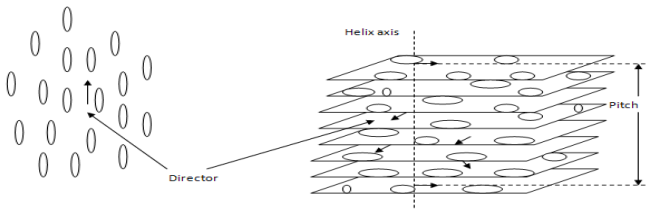

Fig Schematic arrangement in liquid crystal

Buzzer

A buzzer is a small, simple, and widely-used audio signalling device that produces a distinctive buzzing or beeping sound when an electrical current passes through it. It consists of a coil of wire and a diaphragm or vibrating element. When an electric voltage is applied to the buzzer, it causes the coil to generate a magnetic field, which, in turn, attracts the diaphragm or moves it rapidly back and forth. This motion of the diaphragm creates vibrations in the surrounding air, producing the audible sound that we associate with buzzers. Buzzers are commonly employed in various applications, such as alarms, timers, doorbells, and electronic games, to alert users to specific events or conditions due to their simplicity, reliability, and ability to provide immediate auditory feedback.

Fig Buzzer

Power Supplies

A power supply (sometimes known as a power supply unit or PSU) is a device or system that supplies electrical or other types of energy to an output load or group of loads. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

This circuit is a small +5V power supply, which is useful when experimenting with digital electronics. Small inexpensive wall transformers with variable output voltage are available from any electronics shop and supermarket. Those transformers are easily available, but usually their voltage regulation is very poor, which makes then not very usable for digital circuit experimenter unless a better regulation can be achieved in some way. The following circuit is the answer to the problem.

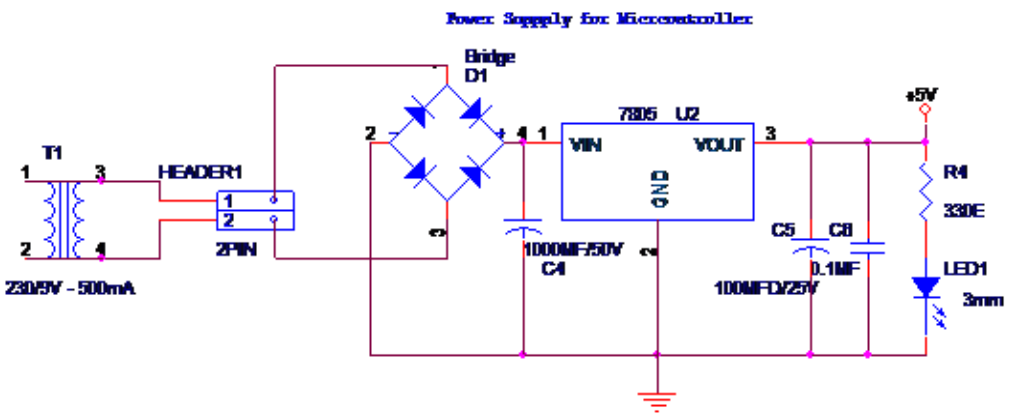

Fig Block diagram of power supply

Transformer

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled wires. A changing current in the first circuit (the primary) creates a changing magnetic field; in turn, this magnetic field induces a changing voltage in the second circuit (the secondary). By adding a load to the secondary circuit, one can make current flow in the transformer, thus transferring energy from one circuit to the other. The secondary induced voltage V_S is scaled from the primary V_P by a factor ideally equal to the ratio of the number of turns of wire in their respective windings:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

By appropriate selection of the numbers of turns, a transformer thus allows an alternating voltage to be stepped up — by making N_S more than N_P or stepped down, by making it less.

A key application of transformers is to reduce the current before transmitting electrical energy over long distances through wires. Most wires have resistance and so dissipate electrical energy at a rate proportional to the square of the current through the wire. By transforming electrical power to a high-voltage, and therefore low-current form for transmission and back again afterwards, transformers enable the economic transmission of power over long

distances. Consequently, transformers have shaped the electricity supply industry, permitting generation to be located remotely from points of demand. All but a fraction of the world's electrical power has passed through a series of transformers by the time it reaches the consumer.

Transformers are some of the most efficient electrical 'machines', with some large units able to transfer 99.75% of their input power to their output. Transformers come in a range of sizes from a thumbnail-sized coupling transformer hidden inside a stage microphone to huge gigavolt-ampere-rated units used to interconnect portions of national power grids. All operate with the same basic principles, though a variety of designs exist to perform specialized roles throughout home and industry.

The transformer is based on two principles: first, that an electric current can produce a magnetic field (electromagnetism) and, second, that a changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). By changing the current in the primary coil, one changes the strength of its magnetic field; since the secondary coil is wrapped around the same magnetic field, a voltage is induced across the secondary.

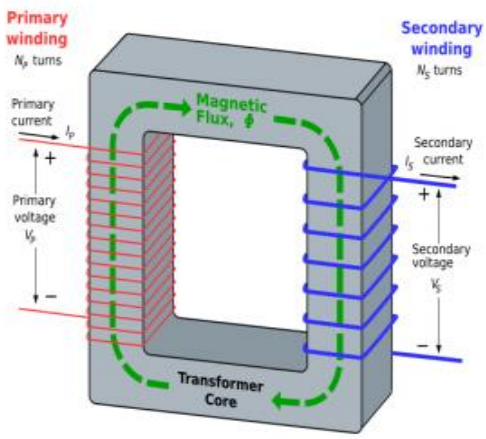

Fig An ideal step-down transformer

A simplified an ideal step-down transformer design is shown in the above figure. A current passing through the primary coil creates a magnetic field. The

primary and secondary coils are wrapped around a core of very high magnetic permeability, such as iron; this ensures that most of the magnetic field lines produced by the primary current are within the iron and pass through the secondary coil as well as the primary coil.

Rectifier

A rectifier is an electrical device that converts alternating current (AC), which periodically reverses direction, to direct current (DC), which flows in only one direction. The process is known as rectification. Rectifiers are used as components of power supplies and as detectors of radio signals. Mainly there are three types of rectifier i.e. half wave rectifier, full wave rectifier and Bridge Rectifier.

Half-wave rectifier

In half-wave rectification of a single-phase supply, either the positive or negative half of the AC wave is passed, while the other half is blocked. Because only one half of the input waveform reaches the output, mean voltage is lower. Half-wave rectification requires a single diode in a single-phase supply, or three in a three-phase supply. Rectifiers yield a unidirectional but pulsating direct current; half-wave rectifiers produce far more ripple than full-wave rectifiers, and much more filtering is needed to eliminate harmonics of the AC frequency from the output.

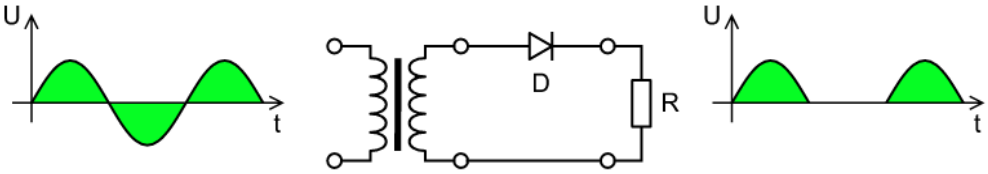

Fig Half Wave Rectifier

Full-wave rectifier

A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to pulsating DC (direct current), and yields a higher average output voltage. Two diodes and a center tapped transformer are needed.

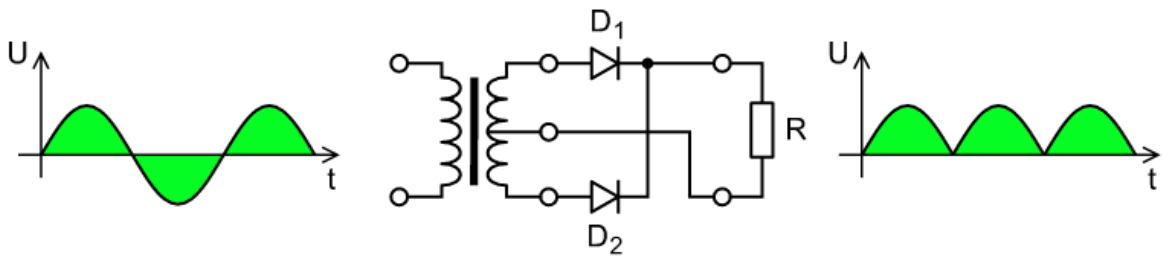

Fig Full-Wave Rectifier

Bridge Rectifier

A diode bridge is an arrangement of four (or more) diodes in a bridge circuit configuration that provides the same polarity of output for either polarity of input. When used in its most common application, for conversion of an alternating current (AC) input into a direct current (DC) output, it is known as a bridge rectifier. A bridge rectifier provides full-wave rectification from a two-wire AC input, resulting in lower cost and weight as compared to a rectifier with a 3-wire input from a transformer with a center-tapped secondary winding. The essential feature of a diode bridge is that the polarity of the output is the same regardless of the polarity at the input.

Basic operation

According to the conventional model of current flow, current is defined to be positive when it flows through electrical conductors from the positive to the negative pole. In actuality, free electrons in a conductor nearly always flow from the negative to the positive pole. In the vast majority of applications, however, the actual direction of current flow is irrelevant. Therefore, in the discussion below the conventional model is retained.

In the diagrams below, when the input connected to the left corner of the diamond is positive, and the input connected to the right corner is negative, current flows from the upper supply terminal to the right along the red (positive) path to the output, and returns to the lower supply terminal via the blue (negative) path.

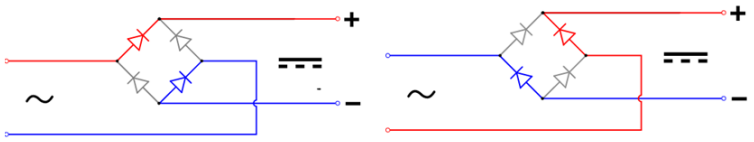

Fig Operation of bridge rectifier

When the input connected to the left corner is negative, and the input connected to the right corner is positive, current flows from the lower supply terminal to the right along the red (positive) path to the output, and returns to the upper supply terminal via the blue (negative) path.

In each case, the upper right output remains positive and lower right output negative. Since this is true whether the input is AC or DC, this circuit not only produces a DC output from an AC input, it can also provide what is sometimes called "reverse polarity protection". That is, it permits normal functioning of DC-powered equipment when batteries have been installed backwards, or when the leads (wires) from a DC power source have been reversed, and protects the equipment from potential damage caused by reverse polarity.

IC Voltage Regulators

Voltage regulators comprise a class of widely used ICs. Regulator IC units contain the circuitry for reference source, comparator amplifier, control device, and overload protection all in a single IC. Although the internal construction of the IC is somewhat different from that described for discrete voltage regulator circuits, the external operation is much the same. IC units

provide regulation of either a fixed positive voltage, a fixed negative voltage, or an adjustable set voltage. A power supply can be built using a transformer connected to the ac supply line to step the ac voltage to desired amplitude, then rectifying that ac voltage, filtering with a capacitor and RC filter, if desired, and finally regulating the dc voltage using an IC regulator. The regulators can be selected for operation with load currents from hundreds of milliamperes to tens of amperes, corresponding to power ratings from milliwatts to tens of watts.

Three-Terminal Voltage Regulators

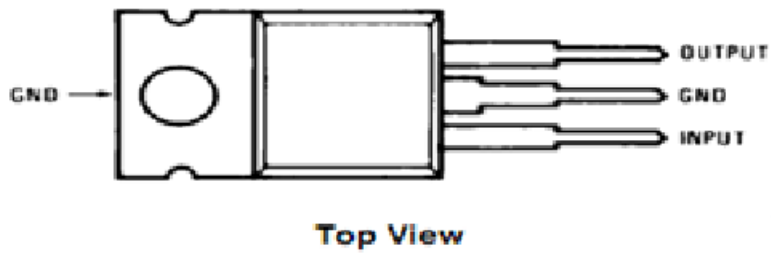

Fig Three-Terminal Voltage Regulators

Figure shows the basic connection of a three-terminal voltage regulator IC to a load. The fixed voltage regulator has an unregulated dc input voltage, V_i, applied to one input terminal, a regulated output dc voltage, V_o, from a second terminal, with the third terminal connected to ground. For a selected regulator, IC device specifications list a voltage range over which the input voltage can vary to maintain a regulated output voltage over a range of load current. The specifications also list the amount of output voltage change resulting from a change in load current (load regulation) or in input voltage (line regulation). The series 78 regulators provide fixed regulated voltages from 5 to 24 V. Figure shows how one such IC, a 7805, is connected to provide voltage regulation with output from this unit of +5V dc. An unregulated input voltage V is filtered by capacitor C1 and connected to the IC's IN terminal. The IC's OUT terminal provides a regulated + 12V which is filtered by capacitor C2 (mostly for any high-frequency noise). The third IC terminal is connected to ground (GND).

While the input voltage may vary over some permissible voltage range, and the output load may vary over some acceptable range, the output voltage remains constant within specified voltage variation limits. These limitations are spelled out in the manufacturer's specification sheets. There are two types of voltage regulator they are 78xx series and 79xx series.

78xx series

There are common configurations for 78xx ICs, including 7805 (5 V), 7806 (6 V), 7808 (8 V), 7809 (9 V), 7810 (10 V), 7812 (12 V), 7815 (15 V), 7818 (18 V), and 7824 (24 V) versions. The 7805 is the most common, as its regulated 5-volt supply provides a convenient power source for most TTL components.

Less common are lower-power versions such as the LM78Mxx series (500 mA) and LM78Lxx series (100 mA) from National Semiconductor. Some devices provide slightly different voltages than usual, such as the LM78L62 (6.2 volts) and LM78L82 (8.2 volts) as well as the STMicroelectronics L78L33ACZ (3.3 volts).

4.4 software requirement

ARDUINO SOFTWARE (IDE)

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.

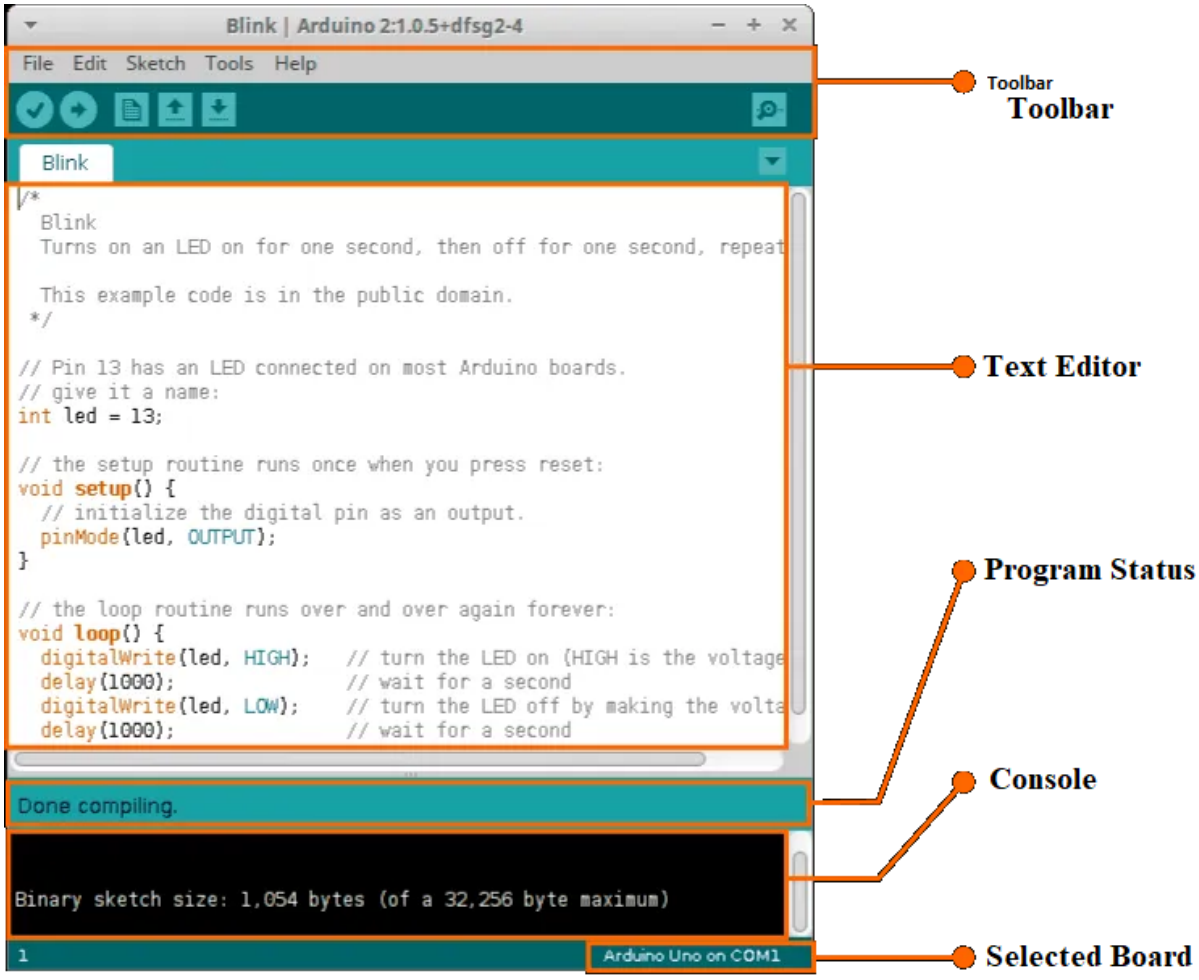

Fig Arduino IDE

- **Writing Sketches**

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

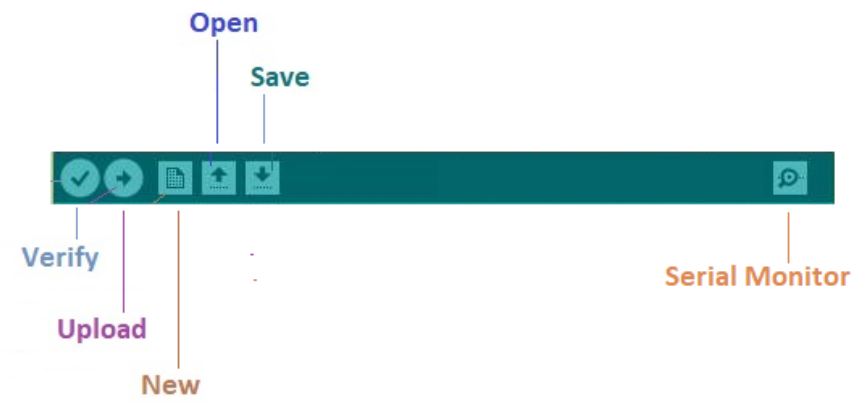

Fig Arduino IDE Tool Icon

- **Verify** Checks your code for errors compiling it.
- **Upload** Compiles your code and uploads it to the configured board. See uploading below for details. Note: If you are using an external programmer with your board, you can hold down the "shift" key on your computer when using this icon. The text will change to "Upload using Programmer"
- **New** Creates a new sketch.
- **Open** Presents a menu of all the sketches in your sketchbook. Clicking one will open it within the current window overwriting its content.
- **Save** Saves your sketch.
- **Serial Monitor** Opens the serial monitor.

Additional commands are found within the five menus: File, Edit, Sketch, Tools, Help. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

File

- **New** Creates a new instance of the editor, with the bare minimum structure of a sketch already in place.
- **Open** Allows to load a sketch file browsing through the computer drives and folders.

- **Open** Recent Provides a short list of the most recent sketches, ready to be opened.
- **Sketchbook** Shows the current sketches within the sketchbook folder structure; clicking on any name opens the corresponding sketch in a new editor instance.
- **Examples** Any example provided by the Arduino Software (IDE) or library shows up in this menu item. All the examples are structured in a tree that allows easy access by topic or library.
- **Close** Closes the instance of the Arduino Software from which it is clicked.
- **Save** Saves the sketch with the current name. If the file hasn't been named before, a name will be provided in a "Save as.." window.
- **Save as...** Allows to save the current sketch with a different name.
- **Page Setup** It shows the Page Setup window for printing.
- **Print** Sends the current sketch to the printer according to the settings defined in Page Setup.
- **Preferences** Opens the Preferences window where some settings of the IDE may be customized, as the language of the IDE interface.
- **Quit** Closes all IDE windows. The same sketches open when Quit was chosen will be automatically reopened the next time you start the IDE.

Edit

- **Undo/Redo** Goes back of one or more steps you did while editing; when you go back, you may go forward with Redo.
- **Cut** Removes the selected text from the editor and places it into the clipboard.
- **Copy** Duplicates the selected text in the editor and places it into the clipboard.

- **Copy for Forum** Copies the code of your sketch to the clipboard in a form suitable for posting to the forum, complete with syntax coloring.
- **Copy as HTML** Copies the code of your sketch to the clipboard as HTML, suitable for embedding in web pages.
- **Paste** Puts the contents of the clipboard at the cursor position, in the editor.
- **Select All** Selects and highlights the whole content of the editor.
- **Comment/Uncomment** Puts or removes the // comment marker at the beginning of each selected line.
- **Increase/Decrease** Indent Adds or subtracts a space at the beginning of each selected line, moving the text one space on the right or eliminating a space at the beginning.
- **Find Opens** the Find and Replace window where you can specify text to search inside the current sketch according to several options.
- **Find Next** Highlights the next occurrence - if any - of the string specified as the search item in the Find window, relative to the cursor position.
- **Find Previous** Highlights the previous occurrence - if any - of the string specified as the search item in the Find window relative to the cursor position.

Sketch

- **Verify/Compile** Checks your sketch for errors compiling it; it will report memory usage for code and variables in the console area.
- **Upload** Compiles and loads the binary file onto the configured board through the configured Port.
- **Upload Using Programmer** This will overwrite the bootloader on the board; you will need to use Tools > Burn Bootloader to restore it and be able to Upload to USB serial port again. However, it allows you to use the full

capacity of the Flash memory for your sketch. Please note that this command will NOT burn the fuses. To do so a Tools -> Burn Bootloader command must be executed.

- **Export** Compiled Binary Saves a .hex file that may be kept as archive or sent to the board using other tools.
- **Show Sketch Folder** Opens the current sketch folder.
- **Include Library** Adds a library to your sketch by inserting #include statements at the start of your code. For more details, see libraries below. Additionally, from this menu item you can access the Library Manager and import new libraries from .zip files.
- Add File... Adds a source file to the sketch (it will be copied from its current location). The new file appears in a new tab in the sketch window. Files can be removed from the sketch using the tab menu accessible clicking on the small triangle icon below the serial monitor one on the right side o the toolbar.

Tools

- **Auto Format** This formats your code nicely: i.e. indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more.
- **Archive Sketch** Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.
- **Fix Encoding & Reload** Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
- **Serial Monitor** Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.
- **Board** Select the board that you're using. See below for descriptions of the various boards.

- **Port** This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu.
- **Programmer** For selecting a hardware programmer when programming a board or chip and not using the onboard USB-serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this.
- **Burn Bootloader** The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino board but is useful if you purchase a new ATmega microcontroller (which normally comes without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the bootloader on the target board. This command also set the right fuses.

Help

Here you find easy access to a number of documents that come with the Arduino Software (IDE). You have access to **Getting Started, Reference**, this guide to the IDE and other documents locally, without an internet connection. The documents are a local copy of the online ones and may link back to our online website.

Find in Reference This is the only interactive function of the Help menu: it directly selects the relevant page in the local copy of the Reference for the function or command under the cursor.

Sketchbook

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the File > Sketchbook menu or from the Open button on the toolbar. The first time you run the Arduino software, it will automatically

create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the Preferences dialog.

- Tabs, Multiple Files, and Compilation

Allows you to manage sketches with more than one file (each of which appears in its own tab). These can be normal Arduino code files (no visible extension), C files (.c extension), C++ files (.cpp), or header files (.h).

Before compiling the sketch, all the normal Arduino code files of the sketch (.ino, .pde) are concatenated into a single file following the order the tabs are shown in. The other file types are left as is.

Uploading

Before uploading your sketch, you need to select the correct items from the Tools > Board and Tools > Port menus. The boards are described below. On the Mac, the serial port is probably something like /dev/tty.usbmodem241 (for an Uno or Mega2560 or Leonardo) or /dev/tty.usbserial-1B1 (for a Duemilanove or earlier USB board), or /dev/tty.USA19QW1b1P1.1 (for a serial board connected with a Keyspan USB-to-Serial adapter). On Windows, it's probably COM1 or COM2 (for a serial board) or COM4, COM5, COM7, or higher (for a USB board) - to find out, you look for USB serial device in the ports section of the Windows Device Manager. On Linux, it should be /dev/ttyACMx , /dev/ttyUSBx or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the Upload item from the Sketch menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as

the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

REVIEW OF LITERATURE

5 REVIEW OF LITERATURE

H Lu, A Ding, Y Zheng et al..,[1] proposed a smart city framework for securing drinking water supply using an early warning system based on an online sensor network and machine learning. The system continuously monitors water quality parameters, including turbidity, pH, and residual chlorine, using IoT-enabled sensors. Machine learning algorithms analyze real-time data to predict potential contamination events. Alerts are generated and communicated to municipal authorities for immediate intervention. The system was tested in multiple urban water supply networks, showing high accuracy in early detection of anomalies. The proposed method reduces reliance on manual inspections and improves public health protection. It also integrates predictive analytics for long-term planning of water safety. Practical applications demonstrated cost-effectiveness and scalability. The system ensures timely intervention to prevent contamination. Overall, the research highlights the effectiveness of IoT and AI for urban water security and management.

BO Adegoke, OF Adegoke et al..,[2] developed a real-time IoT-based water monitoring device with an integrated alert system aimed at improving urban water management. The device measures flow rate, pressure, and water quality indicators and sends data wirelessly to a cloud-based platform. Alerts are triggered when readings exceed predefined thresholds, enabling timely corrective actions. The system was validated in residential and commercial settings, showing reliable detection of leaks and abnormal water consumption. Data visualization dashboards allow stakeholders to monitor trends and identify inefficiencies. Integration with mobile applications ensures alerts reach users and managers promptly. The system enhances operational efficiency while promoting water conservation. IoT connectivity supports remote monitoring and predictive maintenance. The study emphasizes the importance of combining

hardware, software, and communication technologies. Overall, the work contributes to smarter and more sustainable water supply management.

KB Adedeji, AA Ponnle et al..,[3] presented a conceptual model for the digitalization of water supply systems under the Water 4.0 paradigm to support sustainable smart city development. The model integrates IoT sensors, cloud computing, and analytics to monitor and control water distribution in real time. Data-driven insights optimize water usage, detect leaks, and predict equipment failures. Digital twins of water networks are proposed to simulate scenarios and improve decision-making. Pilot implementations demonstrated improved resource management and reduced operational costs. Emphasis is placed on interoperability and scalability for expanding urban infrastructure. The framework addresses water scarcity challenges while promoting smart infrastructure investment. Findings highlight the potential of digital technologies in transforming traditional water utilities. Practical application ensures efficiency and sustainability. Overall, the research provides a roadmap for future smart water systems in urban environments.

R Schneider, K Domakonda et al..,[4] implemented an alert system for communicating actionable wastewater surveillance results to school communities in Houston, Texas (2023–2024). The system collects wastewater samples and analyzes them for pathogens, chemical indicators, and microbial load. Results are processed in real time and sent as alerts to administrators and health officials. This early warning system helps mitigate health risks by informing timely interventions. Pilot deployment in schools demonstrated rapid detection of potential outbreaks. Automated notifications facilitate preventive measures, including sanitation and targeted testing. The integration of IoT sensors and cloud-based reporting improved data reliability and transparency. The study shows the effectiveness of wastewater surveillance in community-level monitoring. Findings suggest such systems can complement traditional

public health strategies. Overall, the research provides a scalable model for early detection in educational or institutional settings.

M Palati, R Dharshan, PK Pattnaik et al..,[5] developed a smart water leakage alert system for residential buildings that combines sensor networks with real-time monitoring algorithms. Flow and pressure sensors detect abnormal water usage patterns indicative of leaks. The system triggers alerts to residents via mobile applications or cloud dashboards. Field testing showed rapid detection of leakage events, reducing water wastage and preventing property damage. Data analytics enables trend analysis and predictive maintenance. The solution enhances water efficiency in urban dwellings and promotes sustainable practices. Integration with existing plumbing infrastructure ensures cost-effective deployment. Reliability and scalability were validated under varying usage conditions. The research demonstrates that automated alerts improve user responsiveness and conservation awareness. Overall, the work contributes to smart home water management technologies.

AD Gupta, P Pandey, A Feijóo, ZM Yaseen, ND Bokde et al..,[6] reviewed smart water technologies for efficient resource management in urban and industrial settings. The study highlights IoT-enabled sensors, data analytics, and predictive algorithms that optimize water distribution, detect leaks, and monitor water quality. The review emphasizes integration with mobile and cloud platforms for remote monitoring and real-time decision-making. Case studies demonstrate reduced water loss, operational costs, and improved efficiency in smart city projects. Challenges include sensor calibration, data reliability, and cybersecurity concerns. The research emphasizes combining real-time monitoring with automated control systems. Future directions include AI-based predictive maintenance and digital twins for enhanced network management. The study underlines environmental and economic benefits of smart water adoption. Findings provide a comprehensive reference for

technology-driven water management strategies. Overall, the review supports sustainable and intelligent water system design.

P Nonthakarn, S Madyod et al..,[7] developed a real-time water monitoring and alert system for sustainable aquaculture in the Tapi River. The system measures water parameters such as temperature, pH, dissolved oxygen, and turbidity using IoT-enabled sensors. Data are transmitted to a cloud platform for real-time analysis and visualization. Alerts are issued to farm managers when parameters exceed safe thresholds, allowing timely interventions. Pilot testing demonstrated improved water quality management and optimized feeding schedules. Integration of IoT, data analytics, and mobile communication ensures operational efficiency. The system reduces environmental risks and resource wastage. Long-term deployment supports sustainable aquaculture practices. Findings highlight practical applications for aquaculture monitoring in rivers and ponds. The proposed system improves productivity and fish health management. Overall, the study demonstrates the potential of smart monitoring in aquatic environments.

HM Ramos, A Kuriqi, M Besharat, E Creaco et al..,[8] explored smart water grids and digital twins for improving efficiency in water distribution networks. The study integrates sensor networks, SCADA systems, and predictive models to optimize operational performance. Digital twins simulate real-time conditions, enabling predictive maintenance and leak detection. Pilot implementations showed reduced water losses and improved pressure management. Data analytics facilitate informed decision-making and resource allocation. IoT integration allows remote monitoring and early anomaly detection. The study emphasizes combining hardware, software, and simulation technologies for sustainable water management. Findings indicate enhanced reliability and reduced operational costs. The approach supports smart city water infrastructure and environmental sustainability. Practical applications

confirm feasibility for large-scale urban deployment. Overall, the research contributes to modernization of water distribution systems.

VA Tzanakakis, NV Paranychianakis et al..,[9] examined water supply challenges and scarcity in urban environments. The study emphasizes the need for smart monitoring systems, efficient distribution networks, and conservation strategies. Sensor-based monitoring enables real-time detection of anomalies such as leaks or pressure drops. Predictive analytics helps forecast consumption patterns and plan resource allocation. Case studies indicate significant reductions in water wastage and improved reliability of supply. Integration with digital twins and IoT platforms supports sustainable urban water management. The study underlines the importance of automated control systems to mitigate scarcity risks. Findings highlight economic, environmental, and social benefits of smart water systems. Practical applications demonstrate enhanced management efficiency and reduced operational costs. Overall, the research provides a framework for sustainable urban water planning.

A Menapace, A Zanfei et al..,[10] addressed burst detection in water distribution systems and the challenge of dataset collection for accurate monitoring. High-resolution sensor networks detect sudden pressure or flow anomalies indicative of bursts. Collected data are used to train machine learning models for predictive maintenance and accurate detection. Pilot studies demonstrated reduced water loss and operational costs compared to traditional methods. The research highlights challenges in dataset availability, labeling, and real-time processing. Integration with SCADA systems improves operational response and decision-making. Findings suggest that automated, data-driven methods outperform manual inspections. The study emphasizes the importance of sensor deployment and analytics for system resilience. Practical applications show improved reliability in water networks. Overall, the work provides insights for effective burst detection and sustainable management.

DEFINITION OF THE PROBLEM AND SCOPE OF THE RESEARCH WORK

6. DEFINITION OF THE PROBLEM AND SCOPE OF THE RESEARCH WORK

6.1 DEFINITION OF THE PROBLEM

Access to sufficient water remains a significant challenge in many rural and semi-urban areas due to inefficient water management systems. Traditional methods rely on fixed schedules and manual monitoring, which often result in water shortages, overflow, and wastage. Leakage in pipelines goes undetected for long periods, causing substantial loss of treated water, while consumers have no real-time information about water availability. Unequal distribution between regions further exacerbates the problem, creating dissatisfaction among residents. The lack of transparency, automated monitoring, and timely communication makes it difficult for authorities to optimize water usage, detect faults, and ensure fair supply. Consequently, there is an urgent need for a system that can provide real-time monitoring, regulate water distribution efficiently, prevent wastage, and enhance sustainability, ensuring that water resources are used judiciously and equitably.

6.2 SCOPE OF THE RESEARCH WORK

The scope of this research work focuses on developing an IoT-based Smart Water Supply Time Scheduler to improve water management in rural villages. It includes monitoring water flow into storage tanks and measuring water levels to provide real-time alerts for availability and overflow conditions. The system aims to automate notifications to villagers, enabling them to plan water usage efficiently, while also regulating supply schedules to ensure equitable distribution across households. Data collected from sensors can be analyzed to identify usage patterns, optimize supply timing, and reduce wastage. The project emphasizes reducing human intervention, enhancing transparency in water distribution, and promoting sustainable practices in resource management.

Furthermore, the system can be extended in the future to integrate remote monitoring by local authorities, predictive maintenance of water infrastructure, and adaptive scheduling based on seasonal demand, making it a scalable solution for improving rural water supply reliability.

ARCHITECTURE DESIGN

7. ARCHITECTURE DESIGN

The system introduces a Smart Water Supply Time Scheduler using IoT to address the inefficiencies and challenges associated with conventional water distribution in rural areas. Traditional water supply systems often face irregular distribution, manual interventions, water wastage, and a lack of transparency, which leads to inconvenience for villagers and mismanagement of resources. To overcome these issues, the proposed system integrates sensors, microcontrollers, communication modules, and scheduling algorithms to create a highly efficient, automated water management network. The architecture design of this system is structured to ensure seamless monitoring, control, and equitable distribution of water while providing timely updates to both villagers and local authorities. The design emphasizes real-time data collection, automated decision-making, and effective communication, forming the backbone of an intelligent and sustainable water supply system.

At the core of the system are water flow sensors and water level sensors strategically installed in overhead or underground storage tanks. These sensors continuously monitor the inflow and outflow of water as well as the current water level within the tanks. The water flow sensor accurately measures the amount of water entering or leaving the tank, providing essential information about water consumption patterns and overall supply. Meanwhile, the water level sensor detects the tank's capacity and generates signals when predefined thresholds are reached. For instance, when the tank is nearing full capacity, an overflow alert is triggered to prevent wastage. Conversely, when water levels drop below a minimum threshold, the system can notify authorities or adjust the supply schedule to maintain uninterrupted water availability. The sensor data forms the foundation of the system, providing real-time information that drives all subsequent processing and decision-making operations.

The microcontroller serves as the central processing unit of the system, receiving data from the sensors and executing control actions based on predefined algorithms. It analyzes the inflow and outflow data to determine water consumption trends, detect anomalies, and make timely decisions regarding water supply. One of the primary functionalities of the microcontroller is its ability to manage the scheduling of water distribution across different households. By implementing a time-based allocation strategy, the system ensures that each household receives water in a fair and organized manner, reducing conflicts and minimizing waiting times. Additionally, the microcontroller logs all sensor data for future analysis, providing a historical record of water usage patterns that can be accessed by local authorities for monitoring, planning, and optimizing supply schedules. This layer of processing ensures that the system is not only reactive but also proactive, capable of predicting potential shortages or overflows based on historical trends.

Communication plays a crucial role in the proposed architecture by connecting the system with end-users and local authorities. Using IoT-enabled communication modules such as GSM, Wi-Fi, or LoRa, the system can transmit real-time notifications and alerts to villagers regarding water availability. When the tank begins filling, villagers receive instant messages on their mobile phones, informing them that water is ready for use. This feature eliminates the need for manual inquiries and waiting in queues, significantly improving convenience and efficiency. Similarly, when the tank reaches maximum capacity, the system generates overflow alerts to prevent water wastage and notify maintenance personnel if required. For authorities, the communication system allows remote monitoring of sensor data, providing transparency and enabling timely interventions in case of anomalies. This connectivity ensures a seamless flow of information between the physical infrastructure and digital

management systems, making the overall water supply process more responsive and accountable.

The architecture also integrates a scheduling module that works closely with the microcontroller to manage the timing of water supply for different households. By allocating water based on a predefined timetable, the system ensures equitable distribution, preventing overuse and conflicts among residents. The scheduling module considers real-time tank levels, consumption patterns, and historical usage data to make informed decisions about supply timing. This automation significantly reduces the need for manual supervision and intervention, which is often prone to errors and inefficiencies in conventional systems. Furthermore, the integration of the scheduling module with the communication system allows villagers to receive notifications about their allocated water timings, ensuring that everyone can access water fairly and without unnecessary waiting.

Data logging and analytics form another critical aspect of the proposed architecture. All sensor readings, supply timings, and consumption data are stored in a cloud-based or local database, providing a comprehensive record of the water management system. This historical data enables local authorities to analyze water usage trends, identify periods of high or low demand, and plan supply schedules more efficiently. It also facilitates the detection of anomalies, such as unexpected drops in water levels or irregular flow rates, allowing for proactive maintenance and intervention. By leveraging IoT analytics, the system not only ensures efficient distribution but also promotes transparency and accountability, as all stakeholders can access verified information regarding water supply and usage.

The architecture is designed for scalability and adaptability, allowing additional tanks, sensors, or households to be integrated into the system as

needed. The modular design ensures that new components can be easily added without disrupting existing operations. Moreover, the system's reliance on IoT technology enables remote monitoring and control, making it suitable for implementation across multiple villages or larger rural networks. The combination of real-time sensing, automated processing, effective communication, and scheduling ensures that water is distributed efficiently, wastage is minimized, and villagers are empowered with timely information. By reducing manual intervention and optimizing resource allocation, the system addresses the core challenges of traditional water supply management and promotes sustainable usage practices.

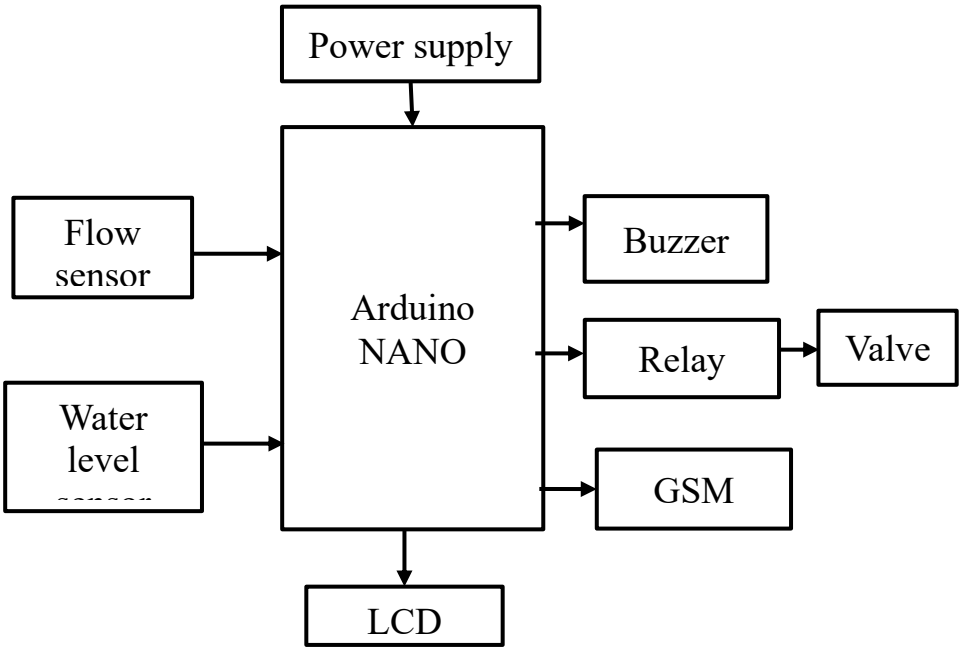

Fig block diagram

RESULT AND DISCUSSION

8. RESULT AND DISCUSSION

The implementation of the IoT-based Smart Water Supply Time Scheduler demonstrated significant improvements in water management for rural villages. During testing, the system effectively monitored real-time water flow and tank levels using sensors, providing accurate data to the central microcontroller. Whenever water entered the storage tank, villagers received instant notifications, allowing them to plan water usage efficiently and reduce unnecessary waiting. The system also successfully detected overflow conditions, triggering alerts that prevented wastage and ensured responsible consumption. Observations from the testing phase showed that the time scheduling module effectively distributed water across multiple households, ensuring fairness and minimizing disputes among residents. The automated notification system reduced dependence on manual monitoring, eliminating errors associated with traditional water supply methods. Data logging functionality proved valuable, providing insights into daily and weekly water usage patterns. These patterns helped in identifying peak demand periods, which can inform future improvements and adjustments in the water supply schedule. Furthermore, the integration of IoT modules enabled seamless communication between the system and end-users, highlighting the potential of smart technologies to enhance rural infrastructure. Users reported increased satisfaction due to timely updates, predictable water supply, and reduced water wastage. The system's ability to operate continuously without human intervention further emphasized its reliability and scalability. Analysis of the collected data also suggested that predictive features, such as anticipating high-demand periods, could be incorporated in future iterations to further optimize supply efficiency. Additionally, the study highlighted that IoT-based systems could play a crucial role in sustainable resource management by reducing wastage, improving transparency, and facilitating better planning. The

discussion indicates that such technologies can serve as a foundation for expanding smart water management across larger areas and integrating additional functionalities such as mobile apps, centralized monitoring, and maintenance alerts. Overall, the results confirm that the proposed system addresses key issues in rural water supply, including wastage, unregulated usage, and inequitable distribution, while offering opportunities for further enhancements through data-driven decision-making and automation. The study demonstrates that IoT-based smart scheduling is a viable and effective solution for sustainable and efficient rural water management.

CONCLUSION AND FUTURE ENHANCEMENT

9. CONCLUSION AND FUTURE ENHANCEMENT

9.1 CONCLUSION

The IoT-based Smart Water Supply Time Scheduler provides a practical solution to the challenges of rural water management. By monitoring water flow and tank levels, it ensures that villagers receive timely notifications about water availability. Overflow alerts prevent wastage and promote efficient use of resources. Scheduled water distribution allows fair access to all households, reducing conflicts and ensuring systematic supply. Real-time monitoring and automated notifications reduce manual efforts and human errors, increasing convenience and reliability. The system also enables data collection and analysis, which can inform future improvements in water management practices. Overall, the proposed system enhances transparency, sustainability, and operational efficiency in rural water supply networks. The results demonstrate that integrating IoT technologies into water distribution can address existing inefficiencies and support better resource utilization. This approach provides a scalable and adaptable solution for villages facing water scarcity and management challenges.

9.2 FUTURE ENHANCEMENT

The system can be enhanced by integrating mobile applications for more user-friendly notifications and scheduling. Predictive analytics could be added to anticipate water demand and optimize supply. Remote monitoring by local authorities can improve management and maintenance. Solar-powered IoT modules can make the system more energy-efficient. Expansion to multiple tanks and water sources can ensure wider coverage. Advanced alerts could app notifications for improved accessibility.

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